

Growth bounds for maximal families of disjoint partitions

A correction of asymptotic fits for OEIS A068598

Report 163

3 October 2026

Abstract

Let $a(n)$ count inclusion-maximal unordered families of strict partitions of n in which no part occurs twice anywhere in the family. We prove

$$\frac{1}{5} \leq \liminf_{n \rightarrow \infty} \frac{\log a(n)}{n \log n} \leq \limsup_{n \rightarrow \infty} \frac{\log a(n)}{n \log n} \leq \frac{1}{4}.$$

The upper bound is elementary and explicit: $a(n) \leq 3^{n-1} n^{\lfloor (n-2)/4 \rfloor}$ for $n \geq 2$. It follows from a recoverable-maximum encoding and the distinct-parts mass inequality credited to Joel Spencer by Nathanson. The lower bound injects ordinary queens configurations into maximal families and uses the published lower bound of Luria and Simkin. A separate elementary construction gives the weaker lower constant $1/7$. Both positive-parameter models $A \exp(Bn^c)$ with $c > 1$ listed in the OEIS entry have ratio $a(n)/(A \exp(Bn^c)) \rightarrow 0$. The first-passage inverse has leading-scale constants between 4 and 5, without assuming monotonicity of $a(n)$. Neither existence of a limiting coefficient, a multiplicative asymptotic equivalent, nor a transseries is established. A bounded exact companion checks conventions and the finite encodings independently of the asymptotic argument.

[write] **Status and trust boundary, added 6 October 2026, batch 107.** This is bundle Report 163 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a068598-disjoint-partition-families** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 163” and its PDF author field is empty. *What rests on what.* The upper bound (1.1) is elementary and self-contained (the mass inequality (2.2), credited to Spencer through Nathanson, is proved where it is used), and so is everything derived from it alone: Corollary 5.1, the upper half of (1.3), the lower constant 4 of (6.1), and the left inequality of Proposition 6.2. The lower constant $1/5$, the linear term of (1.4) and the inverse constant 5 rest on the theorem of Luria and Simkin (Theorem 3.1), which is cited, not reproved; Proposition 4.1 gives the constants $1/7$ and 7 without it. No proof uses a computation. Two statements in OEIS A068598 are corrected, with proofs: Meeussen’s graph comment counts *maximal* cliques (Remark 1.2, which quotes it and works out $n = 8$ in full), and Beregovsky’s fits $a \exp(bn^c)$ with $c > 1$, of 2022 and 2026, are not asymptotic to $a(n)$ (Remark 5.2, which quotes them and derives the failure from the elementary upper bound alone, with explicit ranges of n on which each fit exceeds that bound). Nothing was submitted to the OEIS. The write also adds Remark 3.3 (Simkin’s asymptotic count of queens configurations improves the linear term of (1.4)), Remark 6.3 (the Lambert brackets of Proposition 6.2 against the transseries volume) and Section 9 (open questions, with Proposition 9.1, which locates the families that any proof of the constant $1/4$ would have to count).

Contents

1 The objects and the main conclusion

2

2	An encoding with at most one quarter free labels	5
3	Recoverable bases from queens configurations	7
4	A self contained elementary lower bound	8
5	Why the posted asymptotic fits fail	9
6	First passage inversion without monotonicity	10
7	Finite verification and the reproducible companion	12
8	Source provenance and the remaining gap	13
9	Further questions and research	14

1 The objects and the main conclusion

For an integer $n \geq 1$, let H_n be the finite hypergraph with vertex set $[n] = \{1, \dots, n\}$ and edge set

$$\mathcal{P}_n = \left\{ B \subseteq [n] : \sum_{b \in B} b = n \right\}.$$

An edge is a strict integer partition, represented as a set of positive parts. A family of pairwise disjoint edges is a matching. It is *inclusion-maximal* if no further edge can be added disjointly. The edges and the family itself are unordered. Edge sizes may differ within one family. Let $a(n)$ denote the number of these maximal matchings and set $a(0) = 1$ as the separate sequence convention. Throughout, $\log$ is the natural logarithm.

This is the interpretation of OEIS A068598 supported by its title, examples, and maximal-clique Python code [1]. The graph comment should say *maximal cliques*: counting all cliques would give a different sequence. Maximality does not mean maximum cardinality. For instance the three families at $n = 8$ are

$$\begin{aligned} & \{\{8\}, \{1, 7\}, \{2, 6\}, \{3, 5\}\}, \\ & \{\{8\}, \{1, 2, 5\}\}, \\ & \{\{8\}, \{1, 3, 4\}, \{2, 6\}\}. \end{aligned}$$

Their cardinalities differ. There is no quotient by a symmetry of labels.

Theorem 1.1. *For every integer $n \geq 2$,*

$$a(n) \leq 3^{n-1} n^{\lfloor (n-2)/4 \rfloor}. \quad (1.1)$$

If $Q(k)$ counts ordinary nonattacking placements of k queens on a labeled $k \times k$ board and $k = \lfloor (n-1)/5 \rfloor \geq 1$, then

$$a(n) \geq Q(k). \quad (1.2)$$

Consequently,

$$\frac{1}{5} \leq \liminf_{n \rightarrow \infty} \frac{\log a(n)}{n \log n} \leq \limsup_{n \rightarrow \infty} \frac{\log a(n)}{n \log n} \leq \frac{1}{4}. \quad (1.3)$$

In particular $\log a(n) = \Theta(n \log n)$. More precisely, the lower bound obtained from the queens theorem is the one-sided estimate

$$\log a(n) \geq \frac{n}{5} \log n - \frac{\log 5 + 3}{5} n + o(n). \quad (1.4)$$

The meaning of (1.4) is that some error sequence $r_n = o(n)$ makes the displayed inequality true for all sufficiently large n . It is not an equality or a determination of the linear term of $\log a(n)$. The upper bound is self-contained; the stronger lower constant uses precisely the external theorem stated in Section 3.

Remark 1.2 ([write] The graph comment of A068598, 6 October 2026). The live entry (revision #50, 17 February 2026, read on 6 October 2026; the revision the source inspected) is named “Number of maximal sets of partitions of n with property that all parts in all partitions in the set are distinct.” and carries the comment

Also number of cliques in following graph: each strict partition of n represents a vertex, the relation "having no common integer" defines the edges connecting these. - _Wouter Meeussen_, May 27 2002

(quoted verbatim). Call this graph G_n . A set of vertices of G_n spans a complete subgraph exactly when its strict partitions are pairwise disjoint, that is, when it is a matching of H_n ; so the vertex sets of the *maximal* complete subgraphs of G_n are exactly the maximal matchings, and for $n \geq 1$

$$a(n) = \#\{\text{maximal complete subgraphs of } G_n\}.$$

Read with the now usual meaning of “clique” (any vertex set spanning a complete subgraph), the comment describes a different sequence. *The case $n = 8$ in full.* The strict partitions of 8 are the six vertices

$$v_1 = \{8\}, v_2 = \{1, 7\}, v_3 = \{2, 6\}, v_4 = \{3, 5\}, v_5 = \{1, 2, 5\}, v_6 = \{1, 3, 4\}.$$

The edges of G_8 are the five pairs $v_1 v_j$ ($2 \leq j \leq 6$) and $v_2 v_3, v_2 v_4, v_3 v_4, v_3 v_6$; the other six pairs share a part ($v_2 v_5, v_2 v_6$ and $v_5 v_6$ share 1, $v_3 v_5$ shares 2, $v_4 v_5$ shares 5, $v_4 v_6$ shares 3). Among $v_2, \dots, v_6$ the nonempty complete subgraphs are the five vertices, the four edges and the one triangle $\{v_2, v_3, v_4\}$ (no other triangle exists, since $v_2 v_6$ and $v_4 v_6$ are not edges), ten in all; v_1 is adjacent to every other vertex, so G_8 has $2 \cdot 10 + 1 = 21$ nonempty complete subgraphs (22 with the empty one). The maximal ones are

$$\{v_1, v_2, v_3, v_4\}, \quad \{v_1, v_5\}, \quad \{v_1, v_3, v_6\},$$

three in number, the three families displayed above, and $a(8) = 3$. So G_8 has six vertices, hence at least six nonempty cliques in the usual sense, while $a(8) = 3$. The smallest discrepancy is at $n = 3$: G_3 is one edge $\{3\}\{1, 2\}$, with three nonempty complete subgraphs but one maximal one. For $n = 1, \dots, 12$ the numbers of nonempty complete subgraphs of G_n are 1, 1, 3, 3, 7, 9, 17, 21, 43, 57, 109, 157, against $a(n) = 1, 1, 1, 1, 1, 2, 2, 3, 4, 6, 8, 13$ (the write’s enumeration, whose maximal counts agree with the entry’s data for $1 \leq n \leq 20$). *What is right.* Under the older convention in which a clique is by definition a maximal complete subgraph the comment is correct as written; it is the convention of the Wolfram Language, whose documentation of `FindClique` reads “A clique is a maximal

set of vertices where the corresponding subgraph is a complete graph” [6], and the entry’s three programs all count maximal families: the first Mathematica program tests maximality explicitly, Beregovsky’s Mathematica line uses `FindClique[... ,Infinity,All]`, and Chai Wah Wu’s Python program counts the output of `networkx.find_cliques`, which enumerates maximal cliques. The unambiguous wording is “maximal cliques”, which is the source’s correction. This is recorded here only; nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 107.** *Provenance.* This report is Report 163 of the session bundle that ProveIt’s docs/incoming intake processed as batch 107: the archive `Maximal_Disjoint_Partition_Families_Growth_and_Inverse_Bounds_Source.zip` (444,037 bytes, SHA-256 `f5a5ce9462d5...050163e3`, 15 files at the archive root, no wrapper directory), arrival commit `60f54ea06`, placed in `3988bf5c4`. The text is the delivered `Report163.tex` (565 lines, dated 3 October 2026), shipped as `article.tex`. The delivered nine-page PDF and the checksum manifest `SHA256SUMS` (14 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit (it cites nothing in the repository), no “prepared for private review” line and no personal data; its delivered README used generic sandbox example paths, which the report README does not reproduce. It is a single-source report, so no merge choice was made. The write prefixed the 28 delivered labels with `mdp:` and updated the 20 references to them; it added this subsection, the notes marked [write], Remarks 1.2, 3.3, 5.2 and 6.3, Section 9 and two bibliography entries. No statement, proof or number of the source was changed: the added remarks are the last statements of their sections, the added section comes after the last one, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A068598 [1], live entry at revision #50 (17 February 2026), the revision the source recorded; its 46 data terms ($0 \leq n \leq 45$) are those frozen in `data/source_data.json`, and the linked b-file (Chai Wah Wu, $0 \leq n \leq 46$) adds $a(46) = 4836923613$, which the source did not use. The comments corrected here are quoted in Remarks 1.2 and 5.2.
- The abstracts of Luria and Simkin [3] (arXiv:2105.11431v2) and of Simkin [5] (arXiv:2107.13460v3), and the Wolfram documentation of `FindClique` [6].
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:thm:lambert-core`, `p0:prop:factorial-core`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 6.3.
- Not read by the write: the full texts of Nathanson [2], Luria and Simkin [3], Simkin [5] and Filaseta [4]. The statements made about them in Section 3 and Section 8 are the source’s (for Simkin, the abstract’s); `data/SOURCE_PROVENANCE.json` records what the source inspected, with SHA-256 fingerprints.

What was checked at intake. The batch-107 dossier read the manuscript in full and found no error; it rechecked by hand that every maximal family contains $\{n\}$, the bound $q \leq (n-1)^2/(4n)$ and the floor identity (2.4), the disjointness of the three value classes of (3.2) for $n > 5k$, the injectivity of Lemma 3.2, the greedy product $2^k k!$ for $n > 7k$, the linear coefficient $(\log 5 + 3)/5$ and both Lambert roots of Proposition 6.2, and it verified the two OEIS corrections. It reran the companion on copies (the counts, verification and source outputs reproduce the frozen data). The write rechecked every proof line by line and found no error, recomputed the live OEIS data, enumerated the complete subgraphs of G_n for $n \leq 20$ (Remark 1.2), evaluated the two fits against the upper bound (Remark 5.2), and reran the companion (note at the end of Section 7). Finite computations are observations, not certificates.

Relation to the repository. Before batch 107 no file of the repository mentioned A068598, maximal families of disjoint partitions or counts of queens configurations, and none does outside this report. No statement

of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The Lambert brackets of Section 6 are compared with the transseries volume in Remark 6.3.

[write] **What is not claimed, collected from the source.** The report does not prove that $\lim \log a(n)/(n \log n)$ exists, does not identify its value, gives no multiplicative equivalent and no all-orders expansion or transseries; the linear term in (1.4) belongs to a lower bound only, not to $\log a(n)$. The constant $1/5$ and the inverse constant 5 rest on an external theorem whose proof is not reproduced and which supplies no explicit onset. Monotonicity of $a(n)$ is neither assumed nor asserted, and no exact ceiling formula for $N(y)$ is claimed; floating evaluations of W near an integer certify nothing. Corollary 5.1 rejects the posted fits only as asymptotic equivalents, not as approximations on a finite range. The quadruple-decomposition direction of Section 8 is unproved. The literature check is bounded (Filaseta’s chapter read at abstract level only); the result is described as “an independently verified proof candidate and application of known ingredients, rather than as a certified first proof”. The companion’s finite checks prove no asymptotic statement, and byte identity of rebuilds is toolchain-specific. No OEIS entry was edited. The write adds: Remark 3.3 rests on Simkin’s theorem as stated in its abstract, and the explicit ranges in Remark 5.2 use finitely many floating evaluations of elementary numbers (margins stated there).

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol was renamed; G_n , α and Δ are the write’s).

Symbol	Meaning here (where)	Not to be read as
$a(n)$	maximal families of n , with $a(0) = 1$	the comment’s coefficient a (Remark 5.2)
A, B, c	fit parameters (Section 5); the comments’ a, b, c	the anchor set A and an edge B (Section 2); $A = [1, k]$, $B = [k + 1, k + m]$ and c_i (Proposition 4.1); $B_i(\pi)$ of (3.2)
$H_n, \mathcal{P}_n, G_n$	the hypergraph of strict partitions; its edges; Meeussen’s disjointness graph (Remark 1.2)	a “clique” of G_n in the usual sense (any complete subgraph)
$\mathcal{M}, m, v, q, F, f$	matching without $\{n\}$, its size, labels used, free labels, free set, anchor map (Section 2)	m in Proposition 4.1 (a range length); v in <code>p0:prop:factorial-core</code> (a W_0 value)
$Q(k), k$	ordinary labelled queens configurations; $k = \lfloor (n - 1)/5 \rfloor$ (Section 3), $\lfloor (n - 1)/7 \rfloor$ (Proposition 4.1)	toroidal counts or symmetry classes
π, σ, α	queens permutations; Simkin’s constant 1.942 ± 0.003 (Remark 3.3)	the number π
$N(y)$	$\min\{n \geq 0 : a(n) \geq y\}$	the staircase N_* of <code>p0:def:three-inverses</code> , which presupposes a strictly increasing sequence
L, W, h, r, s	$\log y$; W_0 on $(0, \infty)$; the functions and roots of Proposition 6.2	the target L of <code>p0:prop:factorial-core</code> , which is $L + \log 3$ for r
Δ	excess of the used labels’ sum over $v(v + 1)/2$ (Proposition 9.1)	

2 An encoding with at most one quarter free labels

Every maximal matching contains $\{n\}$. Indeed, no other edge of H_n contains n , since every part is positive. Thus $\{n\}$ could be added to any matching lacking it. Remove this forced edge and write

the remaining matching as $\mathcal{M}$. Let $m = |\mathcal{M}|$ and let v be the number of labels used by its edges. Every remaining edge has at least two parts, so $v \geq 2m$.

For each $B \in \mathcal{M}$, retain its smallest part a_B as an *anchor*, omit its largest part z_B , and call its other parts *free*. Define the disjoint subsets of $[n-1]$

$$A = \{a_B : B \in \mathcal{M}\}, \quad F = \bigcup_{B \in \mathcal{M}} (B \setminus \{a_B, z_B\}),$$

and define $f : F \rightarrow A$ by assigning each free part to the anchor of its edge. Put $q = |F| = v - 2m$.

Lemma 2.1. *The code (A, F, f) determines $\mathcal{M}$ uniquely.*

Proof. For each $a \in A$, recover the corresponding edge as

$$\{a\} \cup f^{-1}(a) \cup \left\{ n - a - \sum_{x \in f^{-1}(a)} x \right\}. \quad (2.1)$$

For a code that came from a matching, the final part is its original largest part. This also recovers two-part edges, whose fibers are empty. The empty matching has the unique empty code. Unused labels need no additional information. Arbitrary codes may fail to reconstruct valid edges or disjoint families; the count below only overcounts by including them. $\square$

The used labels are v distinct positive integers and their sum is mn . Therefore

$$mn \geq 1 + 2 + \cdots + v = \frac{v(v+1)}{2}. \quad (2.2)$$

Nathanson [2, p. 108] explicitly attributes the fixed-arity version of this summation argument to Joel Spencer. We use the same ingredient for variable edge sizes and combine it with the encoding; we make no claim to have originated the mass inequality.

Using (2.2) gives

$$\begin{aligned} q = v - 2m &\leq v - \frac{v(v+1)}{n} \\ &= \frac{(n-1)^2}{4n} - \frac{1}{n} \left(v - \frac{n-1}{2} \right)^2 \leq \frac{(n-1)^2}{4n}. \end{aligned} \quad (2.3)$$

For integers $n \geq 2$,

$$\left\lfloor \frac{(n-1)^2}{4n} \right\rfloor = \left\lfloor \frac{n-2}{4} \right\rfloor. \quad (2.4)$$

Indeed, the left argument equals $(n-2)/4 + 1/(4n)$; the fractional part of $(n-2)/4$ belongs to $\{0, 1/4, 1/2, 3/4\}$ and adding $1/(4n) \leq 1/8$ does not cross the next integer.

There are 3^{n-1} choices of a disjoint pair (A, F) , assigning each label to A , F , or neither. For a pair supporting an actual code, the number of maps is at most $m^q \leq n^q$, and (2.3)–(2.4) bound q . When $A = F = \emptyset$ there is exactly one empty function; when $A = \emptyset$ and $F \neq \emptyset$ there is none. Thus no convention about 0^0 is needed. Lemma 2.1 now proves (1.1), and hence

$$\log a(n) \leq \frac{1}{4}n \log n + (n-1) \log 3. \quad (2.5)$$

Only the initial forced-singleton step used maximality. In particular, the code is also injective on every matching without the singleton.

3 Recoverable bases from queens configurations

A queens configuration on a labeled $k \times k$ board is represented by a permutation π of $[k]$ such that both $i + \pi(i)$ and $i - \pi(i)$ are pairwise distinct. These are ordinary diagonals, with no reduction modulo k , and configurations are not identified under board symmetries. The sole external asymptotic input is the following published theorem.

Theorem 3.1 (Luria and Simkin). *As $k \rightarrow \infty$ through all positive integers,*

$$Q(k) \geq ((1 - o(1))k/e^3)^k. \quad (3.1)$$

The primary version used is Theorem 1.1 on page 1 of arXiv:2105.11431v2 [3]. Its page 2 also explicitly gives $\log Q(k)/(k \log k) \rightarrow 1$, using $Q(k) \leq k!$ for the other direction. The theorem is for ordinary queens through all integers. The use of toroidal placements inside its proof imposes no coprimality condition on the theorem used here. That external proof is not reproduced in this report.

Fix n and $k = \lfloor (n-1)/5 \rfloor \geq 1$. Then $n > 5k$. Given a queens permutation π , define

$$B_i(\pi) = \{i, k + \pi(i), n - k - i - \pi(i)\}, \quad 1 \leq i \leq k. \quad (3.2)$$

Each triple sums to n . Its three value classes lie in

$$[1, k], \quad [k+1, 2k], \quad [n-3k, n-k-2],$$

respectively, and $n-3k > 2k$. Thus the classes are disjoint and each triple is strict. The first two classes have no repetitions because π is a permutation; the third has no repetitions because the sums $i + \pi(i)$ are distinct. The triples therefore form a matching. Only one diagonal condition is needed for this embedding; ordinary queens provide a subset for which the required count is known.

Lemma 3.2. *Choose any maximal extension in H_n of each matching $\{B_i(\pi) : 1 \leq i \leq k\}$. Distinct queens permutations receive distinct extensions, regardless of these choices.*

Proof. A maximal extension exists because the edge set of H_n is finite: add a disjoint edge whenever possible until none remains. Extensions can include edges of all sizes. If a matching contained the bases for both π and σ , then its unique edge containing each $i \in [1, k]$ would equal both $B_i(\pi)$ and $B_i(\sigma)$. Each such edge has exactly one member of $[k+1, 2k]$, so $\pi(i) = \sigma(i)$ for every i . This proves injectivity. $\square$

Lemma 3.2 proves (1.2). The cases $k = 2, 3$, where $Q(k) = 0$, are harmless vacuous inequalities. For large k , taking logarithms in (3.1) yields

$$\log a(n) \geq k \log k - 3k + o(k).$$

Writing $n = 5k + r$ with $r \in \{1, 2, 3, 4, 5\}$ gives

$$k \log k = \frac{n}{5} \log n - \frac{\log 5}{5} n + O(\log n).$$

It follows that (1.4) holds for every sufficiently large n , rather than only a subsequence. Together with (2.5), this completes Theorem 1.1.

Remark 3.3 ([write] Simkin’s asymptotic count, 6 October 2026). After the Luria–Simkin lower bound, Simkin [5] proved that there is a constant $\alpha = 1.942 \pm 3 \cdot 10^{-3}$ with

$$Q(k) = ((1 \pm o(1)) k e^{-\alpha})^k \quad (k \rightarrow \infty)$$

(the abstract of arXiv:2107.13460v3; the write did not read the proof). Using its lower half instead of (3.1) in the computation above gives $\log a(n) \geq k \log k - \alpha k + o(k)$, hence, for all sufficiently large n ,

$$\log a(n) \geq \frac{n}{5} \log n - \frac{\log 5 + \alpha}{5} n + o(n), \quad \frac{\log 5 + \alpha}{5} \in [0.7099, 0.7111],$$

against $(\log 5 + 3)/5 = 0.9219 \dots$ in (1.4). The constant $1/5$ is unchanged: it comes from $k \sim n/5$, not from the size of the queens count. The upper half of Simkin’s theorem bounds only the queens contribution, not $a(n)$. As for (1.4), this is a one-sided estimate, with no explicit onset.

4 A self contained elementary lower bound

The $1/5$ constant depends on Theorem 3.1, but an elementary construction already proves factorial-logarithmic growth.

Proposition 4.1. *If positive integers k, m satisfy $m \geq 2k$ and $3k + 2m < n$, then*

$$a(n) \geq \prod_{i=1}^k (m - 2(i - 1)) \geq (m - 2k + 2)^k. \quad (4.1)$$

In particular, for $n \geq 8$ and $k = \lfloor (n - 1)/7 \rfloor$,

$$a(n) \geq 2^k k!. \quad (4.2)$$

Consequently,

$$\liminf_{n \rightarrow \infty} \frac{\log a(n)}{n \log n} \geq \frac{1}{7} \quad (4.3)$$

without invoking the queens theorem.

Proof. Let $A = [1, k]$ and $B = [k + 1, k + m]$. Choose $b_i \in B$ successively, requiring all b_i and all sums $i + b_i$ to be distinct. At step i , repeating an earlier b_j excludes at most $i - 1$ choices. An equality $i + b_i = j + b_j$ excludes at most another $i - 1$ choices. At least $m - 2(i - 1)$ possibilities remain, proving the product count. Set $c_i = n - i - b_i$. Then

$$c_i \geq n - 2k - m > k + m.$$

The triples $\{i, b_i, c_i\}$ are strict partitions of n with disjoint value ranges. Within those ranges there are no repeated values, the last because the sums $i + b_i$ are distinct. Each base is a matching, and every anchor in A is used. The argument of Lemma 3.2, with middle range B , proves that distinct choices cannot share a maximal extension. This proves (4.1).

Set $m = 2k$ and retain the full product rather than its smallest factor. If $n > 7k$, then

$$\prod_{i=1}^k (2k - 2(i - 1)) = 2^k k!,$$

which proves (4.2) for $k = \lfloor (n-1)/7 \rfloor$. The elementary estimate $\log(k!) = k \log k - k + O(\log k)$ gives

$$\log a(n) \geq \frac{n}{7} \log n + O(n),$$

and proves (4.3). $\square$

Together with (1.1), this gives a wholly elementary proof of $\log a(n) = \Theta(n \log n)$ and of the rejection of the fits below. The factorial bound also provides finite first-passage bounds.

5 Why the posted asymptotic fits fail

The inspected OEIS entry records a 2022 conjectural model with parameters approximately $(A, B, c) = (0.57, 0.062, 1.54)$, and a February 2026 revision to $(0.26, 0.129, 1.36)$ [1]. The following statement rules out either model as an asymptotic equivalent, regardless of its finite-range numerical quality.

Corollary 5.1. *For every fixed $A, B > 0$ and $c > 1$,*

$$\frac{a(n)}{A \exp(Bn^c)} \longrightarrow 0. \quad (5.1)$$

Proof. By (2.5),

$$\log \frac{a(n)}{A \exp(Bn^c)} \leq \frac{1}{4} n \log n + O(n) - Bn^c - \log A \longrightarrow -\infty,$$

since $n \log n = o(n^c)$ for $c > 1$. Exponentiation proves the claim. $\square$

Only the elementary upper bound is needed for this conclusion. It does not imply that the two expressions cannot approximate a short data range. No regression in the companion is used to prove or choose a growth coefficient.

Remark 5.2 ([write] The two posted fits, quoted, and every exponent, 6 October 2026). The live entry (revision #50) carries, verbatim,

Conjecture: $a(n)$ asymptotically grows as $a \cdot \exp(bx^c)$, where $a \approx 0.57$, $b \approx 0.062$, $c \approx 1.54$. - _Elijah Beregovsky_, Nov 12 2022

Better parameter values for the above approximation are $(a, b, c) = (0.26, 0.129, 1.36)$ (from correspondence with _Bill McEachen_). - _Elijah Beregovsky_, Feb 17 2026

(the variable x of the first comment is the index n). Both are of the form $A \exp(Bn^c)$ with $A, B > 0$ and $c > 1$.

(a) *Neither is asymptotic to $a(n)$, nor even logarithmically.* For $n \geq 2$, (1.1) gives $\log a(n) \leq (n-1) \log 3 + \lfloor (n-2)/4 \rfloor \log n \leq n \log 3 + \frac{n}{4} \log n$, so for all $A, B > 0$ and $c > 1$

$$\log \frac{a(n)}{A \exp(Bn^c)} \leq -n^c \left(B - n^{1-c} \log 3 - \frac{1}{4} n^{1-c} \log n \right) - \log A.$$

Since $c > 1$, $n^{1-c} \log n \rightarrow 0$ and $n^{1-c} \rightarrow 0$, so the bracket tends to $B > 0$ and the right side tends to $-\infty$: the ratio tends to 0 (Corollary 5.1). Moreover $\log a(n) \leq (\frac{1}{4} + o(1)) n \log n = o(n^c)$, so

even the weaker reading $\log a(n) \sim Bn^c$ fails. Exponents $c \leq 1$ fail in the other direction: for $n \geq 8$ and $k = \lfloor (n-1)/7 \rfloor$, (4.2) and $k! \geq (k/e)^k$ give $\log a(n) \geq k \log(2k/e) = \frac{n}{7} \log n - O(n)$, so $a(n)/(A \exp(Bn^c)) \rightarrow \infty$ for every $A, B > 0$ and $c \leq 1$. Hence no function $A \exp(Bn^c)$ with $A, B, c > 0$ is asymptotic to $a(n)$; only the upper bound and the elementary Proposition 4.1 are used.

(b) *Explicit ranges.* Let $g(x) = \log A + Bx^c - x \log 3 - \frac{x}{4} \log x$ for $x > 0$. By the first display of (a), $g(n) > 0$ implies $A \exp(Bn^c) > 3^{n-1} n^{\lfloor (n-2)/4 \rfloor} \geq a(n)$. Now $g''(x) = Bc(c-1)x^{c-2} - 1/(4x)$, which is positive for $x > x_c := (4Bc(c-1))^{-1/(c-1)}$; $x_c = 18.60 \dots$ for the 2022 parameters and $45.68 \dots$ for the 2026 parameters. Put $n_0 = 1219$ and $n_0 = 8587$ respectively. Then $n_0 > x_c$, and $g'(n_0) = Bcn_0^{c-1} - \log 3 - \frac{1}{4} \log n_0 - \frac{1}{4}$ equals $1.304 \dots$ and $0.961 \dots$, and $g(n_0)$ equals $0.946 \dots$ and $0.722 \dots$. So $g' > 0$ and $g > 0$ on $[n_0, \infty)$. For the finitely many n with $1215 \leq n < 1219$, respectively $8579 \leq n < 8587$, the direct difference $\log A + Bn^c - \log(3^{n-1} n^{\lfloor (n-2)/4 \rfloor})$ is at least 2.16 (at $n = 1215$), respectively 0.93 (at $n = 8579$). Hence

$$0.57 e^{0.062 n^{1.54}} > a(n) \quad (n \geq 1215), \quad 0.26 e^{0.129 n^{1.36}} > a(n) \quad (n \geq 8579),$$

and both ratios $A \exp(Bn^c)/a(n)$ tend to infinity. The values of g , g' and of the differences are evaluations of explicit elementary numbers in 40-digit floating-point arithmetic (mpmath), with the margins stated; they are not interval-certified. Because of the floor, the comparison with the upper bound is not monotone just below these ranges: the 2022 fit first exceeds it at $n = 1212$ but not at $n = 1214$, and the 2026 fit first at $n = 8576$ but not at $n = 8578$.

(c) *What is not said.* On the data range the fits are fair: the 2022 fit gives 0.487, 0.689 and 0.728 times $a(n)$ at $n = 24, 45$ and 46 , the 2026 fit 0.940, 0.907 and 0.905 times $a(n)$. The second comment calls the formula an approximation; as a finite-range approximation it is not judged here, and the parameter values are not recomputed. What fails is the asymptotic reading of the first comment, which the second inherits: the model class $A \exp(Bn^c)$ cannot describe the growth, which is $\exp(\Theta(n \log n))$ by Theorem 1.1. Nothing was submitted to the OEIS.

6 First passage inversion without monotonicity

For a real threshold $y > 1$ define

$$N(y) = \min\{n \geq 0 : a(n) \geq y\}.$$

The all-integer lower bound implies that this set is nonempty for all $y > 1$. Since each $a(n)$ is finite, $N(y) \rightarrow \infty$ as $y \rightarrow \infty$.

Corollary 6.1. *As $y \rightarrow \infty$,*

$$4 \leq \liminf N(y) \frac{\log \log y}{\log y} \leq \limsup N(y) \frac{\log \log y}{\log y} \leq 5. \quad (6.1)$$

In particular $N(y) = \Theta(\log y / \log \log y)$.

Proof. Write $L = \log y$. Fix $0 < u < 4$ and $w > 5$, and choose $\varepsilon > 0$ so that $(1/4 + \varepsilon)u < 1$ and $(1/5 - \varepsilon)w > 1$. Theorem 1.1 supplies a fixed n_0 such that for all integers $n \geq n_0$,

$$(1/5 - \varepsilon)n \log n \leq \log a(n) \leq (1/4 + \varepsilon)n \log n. \quad (6.2)$$

Let $n_- = \lfloor uL/\log L \rfloor$. For every $n_0 \leq n \leq n_-$, monotonicity of the elementary function $n \log n$ and (6.2) give

$$\log a(n) \leq (1/4 + \varepsilon)n_- \log n_- = (1/4 + \varepsilon)uL(1 + o(1)) < L.$$

The finitely many $n < n_0$ also have $a(n) < y$ for large y . Thus $N(y) > n_-$. Next put $n_+ = \lceil wL/\log L \rceil$. The lower bound in (6.2) gives

$$\log a(n_+) \geq (1/5 - \varepsilon)n_+ \log n_+ = (1/5 - \varepsilon)wL(1 + o(1)) > L,$$

so $N(y) \leq n_+$. Letting u increase to 4 and w decrease to 5 proves (6.1). $\square$

No comparison of $a(n)$ and $a(n+1)$ occurs in this proof. We assert neither monotonicity of the sequence nor an exact inverse equivalent. The inequalities do not justify an exact ceiling formula. If only the elementary lower bound is used, the same proof gives constants 4 and 7 instead.

6.1 Finite elementary inverse bounds

Let W be the positive real Lambert function: $W(t)e^{W(t)} = t$ for $t > 0$. The elementary estimates also give explicit finite bounds.

Proposition 6.2. *For every real $y > 1$, with $L = \log y$,*

$$\left\lceil \frac{4(L + \log 3)}{W(324(L + \log 3))} \right\rceil \leq N(y) \leq 7 \left\lceil \frac{L}{W(2L/e)} \right\rceil + 1. \quad (6.3)$$

Proof. The continuous function $h(x) = x \log(81x)/4$ is strictly increasing for $x \geq 1$, and $h(1) = \log 3$. Its unique root $r > 1$ of $h(r) = L + \log 3$ is

$$r = \frac{4(L + \log 3)}{W(324(L + \log 3))}.$$

By (2.5), every integer $2 \leq n < r$ has $\log a(n) \leq h(n) - \log 3 < L$. The values $n = 0, 1$ separately have $a(n) = 1 < y$. Thus $N(y) \geq \lceil r \rceil$.

For the upper bound let $s = L/W(2L/e)$. Then $s > e/2$ and $s \log(2s/e) = L$. The function $x \log(2x/e)$ is strictly increasing for $x \geq e/2$. With $k = \lceil s \rceil \geq 2$, the elementary integral bound $\log(k!) \geq k \log k - k$ gives

$$2^k k! \geq (2k/e)^k \geq e^L = y.$$

At $n = 7k + 1$, (4.2) therefore yields $a(n) \geq y$, proving the right inequality without monotonicity. $\square$

These are exact-real mathematical inequalities. Uncertified floating-point evaluations of W near an integer do not certify the ceilings. The companion instead uses integer powers and factorials to produce finite brackets at integer thresholds: it excludes every index whose explicit upper bound is too small, and supplies an index $7k + 1$ for which $2^k k!$ reaches the threshold. The finite elementary upper bound has leading constant 7; the asymptotic queens estimate, without an explicit onset here, improves the asymptotic upper constant to 5.

Remark 6.3 ([write] The brackets against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`.

- (a) *The root r is an instance of `p0:prop:factorial-core`, verbatim, with $(\kappa, d, L) := (\frac{1}{4}, \log 3, L + \log 3)$: its phase is $X(\frac{1}{4} \log X + \log 3) = h(X)$, its solution is $v = W_0(4(L + \log 3)e^{4 \log 3}) = W_0(324(L + \log 3))$ and $X = e^{v-4 \log 3} = e^v/81 = 4(L + \log 3)/v = r$ (using $ve^v = 324(L + \log 3)$), and its side condition $\frac{1}{4} \log X + \log 3 > 0$ holds because $r > 1$.*
- (b) *The root s is an instance of `p0:prop:factorial-core`, verbatim, with $(\kappa, d, L) := (1, \log 2 - 1, L)$: the phase is $X(\log X + \log 2 - 1) = X \log(2X/e)$, $v = W_0(2L/e)$ and $X = e^{v+1}/2 = L/v = s$, with side condition $X > e/2$, as in the proof. In both cases the positive real Lambert function of the source is W_0 .*
- (c) *Proposition 6.2 and Corollary 6.1 are not instances of `p0:thm:staircase`, only analogues. $N(y)$ has the form of the staircase $N_*(y)$ of `p0:def:three-inverses`, but that definition presupposes a sequence strictly increasing from some index on, and the theorem an interpolation of it; neither is available, since monotonicity of $a(n)$ is not proved (the data are strictly increasing for $7 \leq n \leq 46$, which proves nothing about later n). The brackets here come from two envelopes, an upper bound excluding small indices and a lower bound at the single index $7\lceil s \rceil + 1$, not from a ceiling of an inverse. If $a(n)$ were proved strictly increasing for $n \geq n_1$, then $N(y) = N_*(y)$ for every $y > \max_{n < n_1} a(n)$, and `p0:thm:staircase(1)–(2)` would apply to any admissible interpolation.*

No novelty is claimed for these inversions.

7 Finite verification and the reproducible companion

The companion uses exact integer arithmetic and the Python standard library. Its production module has no network, input-file, or file-writing API. It prints deterministic JSON and imposes fixed bounds before enumeration. The code is standalone; it does not require any other report or unpublished working file. See the accompanying README for complete commands and output-safety assumptions.

Strict partitions are constructed by subset-sum dynamic programming and checked against a separate recursive generator. A direct increasing- edge matching traversal tests maximality against *every* edge, including earlier edges not chosen. A separate maximal-clique traversal provides another count comparison in a smaller range. Removal of the forced singleton does not change the maximal-family count.

The exact tests cover the following.

1. All values $a(n)$ for $0 \leq n \leq 24$ agree with the inspected OEIS data. The singleton, unordered-family, and inclusion-maximal conventions are checked explicitly.
2. Every matching without its singleton through $n = 24$ is encoded and decoded, including the empty matching and empty fibers. No two matchings have the same code. The mass inequality, the integer free- label bound, and the explicit upper bound are checked exactly.
3. Ordinary queens are enumerated through $k = 8$. For every one of the five choices $n = 5k + r$, $1 \leq r \leq 5$, actual maximal extensions of the triple bases are constructed. Original permutations are recovered from the extensions and distinctness is checked.
4. The upper floor identity and the five-residue inequalities are checked through $n = 100000$. Bounded greedy-construction examples check the choice counts and disjoint triple ranges.

5. Argument validation, workload caps, first-passage scans, intentional fault detection, and byte-identical normal and optimized Python outputs are exercised. No correctness check relies on an `assert` statement that disappears under `-O`.

The first values include

n	16	17	18	19	20	21	22	23	24
$a(n)$	75	115	199	312	533	888	1536	2535	4608.

The threshold command performs an exact bounded scan; failure to reach a threshold only proves that its first passage exceeds the scanned range. It does not extrapolate using the asymptotic bounds. A separate integer bracket command implements the proven finite elementary bounds; its endpoints are not asserted to be exact first-passage indices.

These computations are finite diagnostics, not proofs of the asymptotic statements or of novelty. Earlier independent checks through $n = 22$ and a separate direct-enumeration audit through $n = 24$ served the same diagnostic role. The proof consists of the explicit arguments above together with Theorem 3.1.

The release includes complete LaTeX, a deterministic PDF builder, fixed-allowlist ZIP creation, and SHA-256 checksums. Builds and archive creation use new output names and refuse existing destinations and symlink components; they assume a trusted local parent directory. Two clean PDF builds and two archive constructions are compared byte for byte in the reported environment. Reproduction with a different TeX installation can legitimately change PDF bytes.

[write] **The package here, and the intake reruns, 6 October 2026.** “The release”, “the companion” and “the accompanying README” above are the delivered ones. In this report the companion program and its tests are `code/companion-disjoint_partitions.py` and `code/companion-test_disjoint_partitions.py` (delivered `companion/disjoint_partitions.py` and `companion/test_disjoint_partitions.py`); the frozen outputs are `data/counts.json`, `data/finite_checks.json` and `data/source_data.json`; the companion README is `companion-README.md`; the release tools `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` are in `code/`; `SOURCE_PROVENANCE.json` is in `data/`; the checksum manifest is not shipped. The “earlier independent checks through $n = 22$ ” and the “direct-enumeration audit through $n = 24$ ” were not part of the delivered package. The tests locate the frozen data in `data/` beside the directory holding the program, so they run only on a copy with the delivered names (the report README gives the route); the release tools expect the whole delivered package and POSIX descriptor support, and were not run. On copies at the write (Windows, Python 3.14.4): the `counts`, `verify` and `sources` outputs equal the frozen files once Windows line endings are converted; the threshold scan for 1000 returns $n = 22$ and the integer bracket for 10^{20} is $[26, 127]$; 13 of the 14 companion tests pass, and the one failure compares raw output bytes and sees Windows line endings.

8 Source provenance and the remaining gap

The official A068598 snapshot used here is revision #50, dated 17 February 2026, retrieved from the official OEIS data repository on 3 October 2026. Its definition, displayed terms, and fit comments were also checked against the public entry. A site-wide footer is not treated as an entry revision date. Exact snapshot digests and the inspected source locations appear in the companion’s provenance file; full third-party papers and full entry prose are not redistributed.

Nathanson’s paper [2] was read beyond its abstract. It studies the largest and smallest cardinalities of maximal families when *every* partition has a fixed number of parts. That is a different extremal problem from counting unrestricted maximal families. Its page 108 gives the Spencer attribution

used above. Filaseta’s related 1996 chapter [4] was located and its publisher abstract inspected. That abstract likewise concerns a minimum cardinality at fixed arity. The full subscription chapter was not read, so its arguments and remarks have not been excluded as possible prior art.

Bounded exact-ID and topic searches found no primary source claiming the unrestricted enumeration bounds proved here. Search terms included A068598 with “asymptotic” or “growth”, maximal pairwise disjoint partitions with “number” or “asymptotic”, and disjoint partitions with “ $n \log n$ ”. This is a bounded literature check, not a priority certificate. The result should be described as an independently verified proof candidate and application of known ingredients, rather than as a certified first proof.

The principal mathematical gap is still

$$\frac{1}{5} \leq \liminf \frac{\log a(n)}{n \log n} \leq \limsup \frac{\log a(n)}{n \log n} \leq \frac{1}{4}.$$

The report does not prove that the lower and upper limits coincide, identify their value, establish a multiplicative equivalent, or supply any all-orders expansion or transseries. The linear term in (1.4) belongs only to a lower bound. Improving either counting construction or showing a matching upper and lower coefficient remains open in this treatment.

One concrete unproved direction is to count decompositions of $[4k]$ into k quadruples each summing to $8k + 2$. A bound $k^{2k-o(k)}$ would, by maximal extension and recovery of the used labels, attain coefficient $1/4$ along $n = 8k + 2$. It would not by itself settle all n . Requiring one part from each of four consecutive k -blocks leads to permutations π, σ, τ with $i + \pi(i) + \sigma(i) + \tau(i) = 2k + 2$. The elementary complementary-pair construction gives only $k!$, far short of that proposed count. No stronger count or resulting limit is claimed here.

[write] **The OEIS record and the literature at the commit of this write, 6 October 2026.** The entry was still at revision #50 on 6 October 2026, with the comments quoted in Remarks 1.2 and 5.2, all still present; the b-file term $a(46)$ is noted in Section 1.1. The source does not cite Simkin’s later asymptotic count [5], which sharpens the linear term of the lower bound (Remark 3.3) but not its constant $1/5$. The quadruple-decomposition direction above is restated, with what is missing, as Question 2; Proposition 9.1 explains why it is the natural one.

9 Further questions and research

[write] **Added 6 October 2026, batch 107.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The source’s Section 8 names the gap and one unproved direction; its non-claims are collected in Section 1.1. The intake found no false claim in the source and the write found none either; the two corrections concern OEIS text (Remarks 1.2 and 5.2). The write proves one structural statement first.

For a maximal family of $n \geq 2$ let $\mathcal{M}$, m , v , q be as in Section 2, and let

$$\Delta = \sum_{B \in \mathcal{M}} \sum_{b \in B} b - \frac{v(v+1)}{2} = mn - \frac{v(v+1)}{2},$$

the excess of the sum of the used labels over its least possible value ($\Delta \geq 0$ by (2.2)).

Proposition 9.1 ([write] Where the constant $1/4$ would have to come from). (1) *Exactly*, $q =$

$$\frac{(n-1)^2}{4n} - \frac{1}{n} \left(v - \frac{n-1}{2} \right)^2 - \frac{2\Delta}{n}.$$

- (2) For real $t \geq 0$ let $a_t(n)$ be the number of maximal families of n with $(v - \frac{n-1}{2})^2 + 2\Delta \geq tn^2$. Then $a_t(n) \leq 3^{n-1}n^{(1/4-t)n}$ for every $n \geq 2$.
- (3) For every fixed $t > 1/20$, $a_t(n)/a(n) \rightarrow 0$.
- (4) If $\log a(n_j)/(n_j \log n_j) \rightarrow 1/4$ along a sequence $n_j \rightarrow \infty$, then $a_t(n_j)/a(n_j) \rightarrow 0$ for every fixed $t > 0$. Hence, for every $\varepsilon > 0$, all but a vanishing proportion of the maximal families of $n = n_j$ have $|v - \frac{n-1}{2}| \leq \varepsilon n$ and $\Delta \leq \varepsilon n^2$, and these have $m = n/8 + O(\varepsilon n) + O(1)$ non-singleton edges, of average size $v/m = 4 + O(\varepsilon)$ for ε small.

Proof. (1) Since $mn = v(v+1)/2 + \Delta$, $q = v - 2m = v - v(v+1)/n - 2\Delta/n$, and $v - v(v+1)/n = (n-1)^2/(4n) - \frac{1}{n}(v - \frac{n-1}{2})^2$ as in (2.3). (2) For the families counted, (1) gives $q \leq (n-1)^2/(4n) - tn \leq (1/4 - t)n$. The counting of Section 2 applies unchanged: at most 3^{n-1} pairs (A, F) , and for each at most $n^q \leq n^{(1/4-t)n}$ maps, counted as there (when $(1/4 - t)n < 0$ no family qualifies, since $q \geq 0$); Lemma 2.1 makes the code injective. (3) By the left inequality of (1.3), $\log a(n) \geq (\frac{1}{5} - o(1))n \log n$, while by (2) $\log a_t(n) \leq (\frac{1}{4} - t)n \log n + n \log 3$; the difference is at most $-(t - \frac{1}{20} - o(1))n \log n + n \log 3 \rightarrow -\infty$. (4) The same computation with $\log a(n_j) \geq (\frac{1}{4} - o(1))n_j \log n_j$ gives $a_t(n_j)/a(n_j) \rightarrow 0$ for each $t > 0$. A family not counted by a_{ε^2} has $|v - \frac{n-1}{2}| < \varepsilon n$ and $2\Delta < \varepsilon^2 n^2$; for $\varepsilon \leq 1$ this gives the two stated bounds. Then $v = n/2 + O(\varepsilon n) + O(1)$, $m = (v(v+1)/2 + \Delta)/n = n/8 + O(\varepsilon n) + O(1)$, and $v/m = 4 + O(\varepsilon)$ once n is large. $\square$

Part (3) uses Theorem 3.1 through (1.3); with the elementary (4.3) alone it holds for $t > 3/28$. Part (4) says that a proof of $\limsup \log a(n)/(n \log n) = 1/4$ must count families whose non-singleton edges use about $n/2$ labels, with a nearly minimal label sum and average edge size about 4: the regime of Question 2. It proves nothing about whether such families are numerous.

1. **The limiting coefficient** (source: Section 8, “The report does not prove that the lower and upper limits coincide, identify their value, establish a multiplicative equivalent, or supply any all-orders expansion or transseries”). Proved: $\frac{1}{5} \leq \liminf \leq \limsup \leq \frac{1}{4}$ for $\log a(n)/(n \log n)$. Open: whether the limit exists, and its value; a multiplicative equivalent, or at least the linear term of $\log a(n)$ (only lower estimates are known, (1.4) and Remark 3.3). Missing: a construction beating the queens injection, or an upper bound sharper than the free-label count. Proposition 9.1 says where an improvement of the lower bound to $1/4$ would have to come from.
2. **Quadruple decompositions** (source: Section 8, “One concrete unproved direction”). Let $D(k)$ be the number of decompositions of $[4k]$ into k quadruples each summing to $8k+2$. The source’s implication is correct: *[write] if $D(k) \geq k^{2k-o(k)}$, then $\log a(n) \geq (\frac{1}{4} - o(1))n \log n$ along $n = 8k+2$.* Proof: such a decomposition is a matching of H_{8k+2} (its quadruples are disjoint and sum to n , and $4k(4k+1)/2 = k(8k+2)$ is consistent); extend it maximally. In the extension the edges meeting $[4k]$ are exactly the k quadruples, because every label of $[4k]$ lies in one of them and edges are disjoint; so the decomposition is recovered and distinct decompositions give distinct maximal families. Hence $a(8k+2) \geq D(k)$ and $\log a(n) \geq (2k - o(k)) \log k = (\frac{1}{4} - o(1))n \log n$. Open: whether $D(k) \geq k^{2k-o(k)}$; the source’s block construction (one part from each of four consecutive k -blocks, permutations π, σ, τ with $i + \pi(i) + \sigma(i) + \tau(i) = 2k+2$) and its complementary-pair construction give only $k!$. Even a positive answer settles only $n \equiv 2 \pmod{8}$, not the full limit.

3. **Effective onsets and monotonicity** (source: Section 6, “the asymptotic queens estimate, without an explicit onset here, improves the asymptotic upper constant to 5”, and the companion README). The elementary bounds are explicit for every n and every $y > 1$ (Proposition 6.2); the queens-based constants $1/5$ and 5 have no onset, because the $o(1)$ of Theorem 3.1 is not made explicit. Open: an explicit k_0 and error function in a queens lower bound, hence explicit forms of (1.4) and (6.1). Added by the write (not a claim of the source): is $a(n)$ nondecreasing, and strictly increasing for $n \geq 7$? The data are, through $n = 46$; a proof would make $N(y)$ a staircase in the sense of Remark 6.3(c). Missing: any comparison between maximal families of n and of $n + 1$.
4. **The literature boundary** (source, Section 8). Filaseta’s chapter [4] was read at abstract level only, and the search was bounded. Open: whether it, or other work on maximal families of disjoint partitions or on maximal matchings of the hypergraph H_n , already counts these families. Missing: reading it.

References

- [1] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A068598, entry by Naohiro Nomoto with later contributions. Revision #50, 17 February 2026; inspected 3 October 2026. oeis.org/A068598. Official data: A068598.seq.
- [2] M. B. Nathanson, *Largest and smallest maximal sets of pairwise disjoint partitions*, Journal of Number Theory **17** (1983), 103–112. Spencer’s summation argument is on p. 108. Author-hosted primary PDF.
- [3] Z. Luria and M. Simkin, *A lower bound for the n -queens problem*, Proceedings of the 2022 Annual ACM–SIAM Symposium on Discrete Algorithms (SODA), 2022. doi:10.1137/1.9781611977073.86. Version inspected: arXiv:2105.11431v2 (9 July 2021), Theorem 1.1 on PDF p. 1 and consequence on p. 2.
- [4] M. Filaseta, *The smallest maximal set of pairwise disjoint partitions*, in D. V. Chudnovsky, G. V. Chudnovsky, and M. B. Nathanson (eds.), *Number Theory: New York Seminar 1991–1995*, Springer, 1996, 103–113. doi:10.1007/978-1-4612-2418-1_8. Publisher metadata and abstract were inspected. The full chapter was not read.
- [5] M. Simkin, *The number of n -queens configurations*, Advances in Mathematics **427** (2023), 109127. doi:10.1016/j.aim.2023.109127. Version read by the ProveIt write (abstract only): arXiv:2107.13460v3 (23 November 2022).
- [6] Wolfram Research, *FindClique*, Wolfram Language documentation, read by the ProveIt write on 6 October 2026. Documentation page.